

Evolution and Culture:

A Story of Fear



The Lovers II, 1928 by Rene Magritte

12.16.2024

PSYC5562 Theories of Human Uniqueness

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Abstract

Fear is commonly perceived as a negative emotional state or protective defense mechanism, but is that all? This literature review attempts to peel back the curtain on fear, examining “*metafear*,” the fear of fear itself, as a primary driver of cultural and evolutionary history. By examining disciplines from evolutionary biology to cultural anthropology, the review reveals fear as more than a survival mechanism: it is a powerful force shaping human social dynamics, cognitive processes, and cultural practices. Starting from an evolutionary and neurological perspective, the paper argues that fear catalyzes social and biological action, serving as a valuable precondition to cooperation, curiosity, and reproduction. The research emphasizes fear’s paradoxical nature, demonstrating how it simultaneously fosters social bonding and triggers defensive aggression. Through empirical research, the review shows how fear influences neurological architecture, reproductive strategies, and social information processing. Like any human trait, fear is not without drawbacks. The review communicates the vulnerabilities humans face, showing how fear, primarily through cultural and religious institutions, can lead to subjugation and destructive competition. Examining fear from non-human to human contexts, the research ultimately seeks to understand this emotion at its most raw philosophical level, while pointing toward human-technology interaction as a critical arena for future investigation into the nature of fear.

Introduction

What is fear? Is it an anticipation of evil, a protective response, a simple emotional state, or a feeling of respect? Perhaps it is all of the above. One thing is certain: fear is an enduring part of the human condition, continuing to shape us in a multitude of ways. Often viewed negatively, as something to overcome or avoid, fear may, in fact, be the catalyst for some of humanity's greatest achievements, particularly in the realms of cooperation and innovation.

While *Dune* famously states, “fear is the mind killer,” what happens when we fear fear itself? Can this “*metafear*” be reframed as courage, a virtue highly valued across civilizations and central to human evolution and culture?

This literature review seeks to explore these questions by examining fear as a driving force for cooperation and innovation, not only among humans but also in non-human animals. Rather than focusing on human uniqueness, the aim is to understand how human fear differs from that of other species and how it impacts us in ways that are distinct from our non-human counterparts.

The objective of this review is to challenge the common tendency to villainize fear as merely a negative psychological state. Instead, it will present fear as a necessary and beneficial force, one that, despite its potential drawbacks, plays a critical role in our evolution and society. Fear's transformation into a socially constructive force through abstract thinking is what sets humans apart from other species. Our conceptions of the unknown, of nature, and of life and death drive us to seek control over ourselves and our environments, leading to sophisticated social cooperation, technological innovation, and collective action, phenomena not seen to the same degree in non-human animals.

The Evolutionary Paradox of Human Fear

Human fear is one of the most complex and misunderstood psychological phenomena in our evolutionary history. Rather than being a simple survival mechanism, fear functions as an adaptive system that has shaped human social, cognitive, and cultural development. The traditional view of fear as an isolated defensive response to immediate threats overlooks its larger encompassing role in human evolution and social dynamics. Fear is not just a driver of protective behavior but also promotes cooperation, learning, and social cohesion, common products of human “uniqueness”, survival, and success.

Grossmann’s (2022) “fearful ape hypothesis” presents the primate as a child of fear, challenging the common belief that fear is a purely reactive, maladaptive trait, arguing that humans are more fearful than their closest primate relatives. This idea calls for a reevaluation of fear’s adaptive functions, suggesting that increased fear may have played a significant role in the development of higher cognitive and social abilities in humans. The hypothesis extends far and wide, implicating all our understandings of “being human.” It encourages readers to conceive of fear as more than a survival mechanism, also acting as a positive agent, fostering problem-solving, more intricate social structures, and the creation of culture.

Empirical evidence from Herrmann et al. (2011) supports Grossmann’s hypothesis, showing a distinct behavioral difference between humans and other primates. While chimpanzees, orangutans, and bonobos are curious or indifferent to new stimuli, human infants tend to avoid them. This response is not a minor behavioral difference but reflects significant neurological and evolutionary adaptations. The human tendency to avoid potentially dangerous situations, rather than explore them, suggests an evolutionary shift that prioritized social

cohesion and safety over unrestrained exploration. Although this caution might seem maladaptive compared to the risk-taking behaviors of other primates, it likely provided advantages in fostering cooperation and collective action.

This shift is linked to the development of more sophisticated cognitive and social systems in humans. The neurochemical profile of human fear, characterized by higher levels of dopamine, serotonin, and oxytocin, indicates that fear is more involved in social bonding and cooperation than initially thought. According to Hirter et al. (2021), this neurochemical structure helps regulate both individual and collective behaviors, promoting empathy and trust within social groups. Raghanti et al. (2018) further highlight the evolutionary advantages of these neurochemical changes, noting that higher oxytocin levels in response to fear signals strengthen social bonds, especially in cooperative child-rearing practices.

These adaptations are closely tied to the evolution of human social structures. As humans transitioned from small, isolated groups to larger, more elaborate societies, the mechanisms that govern fear had to evolve to support collaboration over competition. Unlike the largely solitary and territorial behaviors observed in primates, early human communities faced the challenge of aligning individual interests to ensure survival. The heightened sensitivity to threats, influenced by the neurochemical makeup of fear, mediated this transition, making cooperation not just advantageous but essential. In this context, fear became a tool for regulating social behavior, encouraging conformity, increasing tolerance, and maintaining collective security through vigilance against external and internal threats.

Neurological Architecture: The Complex Neural Landscape of Fear

The human brain's fear processing system is a remarkable product of evolution. Unlike

the simpler threat-detection systems found in other species, the neural architecture behind human fear involves many interconnected networks that engage multiple brain regions in response to various threats. With this, humans are able to process fear not just as a reflex to immediate danger but as an adaptive tool for dealing with social, emotional, and environmental challenges. Danger being a purely subjective concept, there is no limit to its semantic qualities and therefore, it doesn't go too far to assume an irrevocable link between fear and reasoning within humans.

Adolphs' (2013) research highlights the interconnected nature of these neural systems, showing that fear processing involves perception-action interactions within inferior frontal brain regions. This suggests that fear is more than just a reflex; it deals with high-level processing. Fear not only triggers bodily responses but also activates cognitive functions such as attention, memory, and social processing. Fear, therefore, is a psychological experience that integrates both physiological and cognitive components, coordinating various mental and emotional faculties. The inferior frontal cortex, amygdala, and prefrontal cortex work together to ensure that individuals not only react to threats but also consider past experiences and future consequences.

The amygdala, often seen as the brain's primary center for fear, plays a central role within a larger neural network. While it is involved in threat detection and emotional processing, it is not an isolated part of our brains. The amygdala also participates in threat assessment, social responses, and emotional memory regulation (LeDoux, 2012). Its interaction with the prefrontal cortex, which governs executive functions like decision-making and impulse control, complicates our understanding of fear. Rather than following a simple linear path, the amygdala's role is part of a dynamic feedback loop that continuously reassesses perceived danger and integrates social cues. It lives at the heart of how we define our realities.

Genetic variations in the dopamine system also influence fear responses, highlighting the idea that fear is shaped by both innate processes and individual life experiences (Grossmann et al., 2011). This modulation shows that fear is not fixed; it adapts based on genetic, experiential, and environmental factors. The dopamine system, which regulates motivation, reward, and mood, can intensify or reduce fear responses, affecting how individuals perceive and react to threats. Dopamine is also involved in reinforcement learning, helping the brain link specific cues to fear over time. This neuroplasticity makes fear a flexible mechanism, evolving through the interaction of genetic predisposition, personal experience, and context.

The hippocampus also takes part in fear by integrating past experiences with current contexts. It helps differentiate between safe and dangerous situations, shaping the emotional response to threats. Unlike other species that may react reflexively to stimuli, humans assess threats through self-cooperation (metacognition), drawing on memory and learned associations to adjust their responses. The communion taking place in the brain among these distinguishable regions, exemplifies the adaptability of human fear, one that concerns itself with the upper echelons of social and biological life more than in other animals.

Fear as a Cooperative Social Strategy: Beyond Survival

In continuing with the theme, recent research discusses fears' significant role in promoting social cooperation. Studies on child development place fear at the center of social behaviors and attachment. Rajhans et al. (2016) found that early fearfulness in children is linked to prosocial behaviors like empathy and sharing. Initially, this comes off as contradictory but digging deeper, fear's role in encouraging cooperation and strengthening group ties becomes more apparent. Fear responses in children help form social bonds, enabling them to engage in

social environments more effectively.

Further evidence comes from Peltola et al. (2015), who showed that fear responses in children predict secure attachment patterns, which are vital for emotional and social development. Secure attachment ensures that children remain connected to their caregivers and social groups, particularly when they are most vulnerable. Fear, triggered by potential separation from caregivers or unfamiliar settings, helps children form the stable relationships needed for their survival and well-being.

Kochanska et al. (2002) expanded on this idea, showing that fear is linked to increased guilt and rule-following behaviors. Fear motivates individuals to adhere to social rules and internalize them, becoming part of their social identity. This process helps people align their behavior with group norms, ensuring that individuals contribute to collective goals. Fear, in this case, serves as a mechanism that guides behavior for the benefit of both the individual and the group.

Fear's role in social strategies also extends to broader group dynamics. In group-living species like humans, fear helps maintain social hierarchies and cooperation. Kossowska (2023) notes that social fear and trust are closely linked, with fear signaling the need to align with group norms. By promoting behaviors that reduce conflict and increase cooperation, fear stabilizes social structures. Research in evolutionary psychology shows that individuals who are more attuned to the emotions of others and who regulate their fear responses appropriately tend to have higher social standing and stronger bonds within their communities (Goetz et al., 2010). Fear is not just about avoiding immediate threats but is woven into the social fabric that supports group cohesion and the collective well-being of the group.

The Reproductive Value of Anxiety

Jacobson and Roche's (2018) research challenges traditional views on anxiety as anti-social instead presenting another paradoxical idea: that anxiety is positively related with reproductive success. Their study, which examined a three-generational sample, found that individuals with moderate to extreme anxiety demonstrated increased fertility across generations. This raises the question once again, why do we fear? Why do we become anxious? For nothing exists as purely negative. And every negative quality necessitates an equally positive quality in return. Jacobson and Roche's research beautifully depicts this principle, reframing anxiety as a trait that may have evolved to optimize survival and reproduction. Moderate anxiety could promote heightened vigilance, cautious decision-making, and an improved ability to anticipate and avoid threats, traits that would have been beneficial in ancestral environments where threat detection was essential. This caution could lead to decisions that protect both the individual and their offspring, potentially enhancing reproductive success.

When balanced, anxiety may have promoted adaptive behaviors, such as conserving resources, increasing awareness of environmental cues, and fostering careful social interactions. Individuals with higher anxiety may have been more likely to avoid risky behaviors, thus improving their chances of survival and reproductive success. This aligns with evolutionary theories suggesting that emotions like anxiety are part of an adaptive toolkit for managing uncertainty and navigating complex environments (Mermelstein et al., 2022). Anxiety-driven hypervigilance could have helped individuals avoid predators or other environmental threats, protecting their own lives and those of their offspring.

Further evidence supporting this view comes from recent research by Yuan et al. (2025)

who show that anxiety is linked to greater sensitivity to social cues and increased engagement in social situations. These behaviors could lead to stronger social bonds and improved outcomes for offspring through more protective and attentive care. Individuals with higher anxiety may be more responsive to the emotional needs of others, fostering closer relationships that enhance group cohesion and collective survival. This ability to detect subtle social signals would have helped individuals navigate complex social environments, improving reproductive success within cooperative groups.

Curiosity, Fear, and Social Information Processing

Another fascinating and once again, seemingly contradictory product of fear is curiosity and the drive to explore, the psychological precondition for much of human cultural progress. Fitri et al.'s (2020) research found a strong positive correlation between death anxiety and social information-seeking behaviors. Fear, rather than merely causing avoidance or inaction, can be a driver for actively seeking information, especially in social contexts. This ability to process and transform fearful experiences into social learning enhances individual survival and contributes to collective intelligence, supporting societal adaptation.

Building on the work of Trudewind (2000) and Litman and Pezzo (2007), this research shows that anxiety can stimulate curiosity, motivating individuals to gather information and solve problems. Humans seem to have evolved a mechanism for turning psychological threats into opportunities for learning and social understanding. Fear, in this sense, does not only protect individuals but also encourages the pursuit of knowledge that can help them respond to social challenges. Intuitively it makes sense; to conquer our fears we must venture out of our “comfort zone” which is a fancy way to say, we must be curious. Therefore, the initial fear gives birth to

our willingness to seek knowledge with the “*metafear*” (fear of fear) mediating the transition.

This ability to be curious under threat is linked to the brain’s activation of neural circuits involved in both fear processing and exploration. According to Loewenstein (1994), curiosity arises from the need to resolve uncertainty, which overlaps with the brain’s response to fear. When faced with a threat, the brain doesn’t only trigger a fight-or-flight response, it also activates regions responsible for seeking out information that could help mitigate or understand the source of fear. This interaction between fear and curiosity forms an adaptive mechanism, where psychological discomfort prompts proactive social learning behaviors rather than retreat.

Cultural and Religious Manifestations of Fear

It goes without saying, fear’s existence as an individual phenomenon implicates collective cultural processes and practices as well. Norenzayan and Shariff’s (2008) research on religious prosociality reveals how fear mechanisms contribute to group cohesion and behavioral norms. Their findings show that religious thoughts, when experimentally induced, can reduce cheating and increase altruistic actions, even among anonymous individuals. This suggests that fear-based cultural mechanisms are effective tools for maintaining social coherence and promoting collective moral standards. Fear in religion, therefore, serves not only as a response to divine punishment or the afterlife but also as a social force that reinforces moral behavior and supports societal stability.

The relationship between fear and religious behavior is not simple, as religious ideologies often address deep-seated anxieties about mortality, suffering, and the unknown. Becker (1973) argues that the fear of death (unknown) is a fundamental motivator behind religious and cultural systems. Belief in an afterlife or divine justice offers a framework for managing existential fears.

Religious doctrines provide explanations for suffering and paths to transcendence, mitigating the psychological impact of fear by offering hope, meaning, and control over life's uncertainties. In this way, religious rituals, commandments, and moral codes are also born from the fire of curiosity, attempting to regulate the fear of nature and the unknown which results in large-scale group cohesion.

Cultural mechanisms tied to fear help structure societies around shared anxieties, promoting conformity and a sense of belonging. Fear of social exclusion or divine punishment, for example, may encourage adherence to moral codes and prosocial behavior, thus benefiting group survival. When individuals fear punishment, whether divine or social, they are more likely to conform to group norms. Fear-based cultural practices and religious rituals, therefore, not only respond to existential fears but also act as adaptive social strategies for maintaining the functioning of complex societies.

However, discussed in detail in many forms of media and art, the relationship between fear and culture is not without its drawbacks. Bertrand Russell's (1927) critical perspective offers a counterpoint, suggesting that while religious institutions arise from fear, they can also foster destructive social dynamics. He argues that religious doctrines, often based on fear of the unknown or divine wrath, can create rigid social structures that limit freedom and critical thought. According to Russell, fear-driven religious systems can divide groups, promote intolerance, and justify violence in the name of divine will. While fear can create social cohesion, it can also perpetuate fear, fear of difference, outsiders, and change.

Russell's view highlights the potential for (religious) institutions to manipulate fear for control, using it as a tool to shape behavior. This use of fear can hinder intellectual progress and

social evolution by discouraging questioning and fostering dogma. The emphasis on faith over reason, for example, can keep individuals in a state of fear and subjugation, stifling intellectual freedom and limiting societal growth. Religious and cultural institutions, then, act as double-edged swords: they can provide security and meaning but also create environments of control and oppression through the continued manipulation of fear. It follows the blessing and the curse that come with enhanced cognition and communication through the mind, body, and environment. Phenomena like fear can drive historical progress both in the most enlightening and tragic way.

Competitive and Defensive Dimensions of Fear

Evident from the previous section, the social impact of fear is not always positive. In reality, fear can weaken social bonds, incite hostility, and disrupt harmony. Halevy's (2023) research discusses precisely this, delineating fear to discourage collaboration, prompt defensive aggression, and motivate power-seeking behaviors.

Fear is a strong motivator of defensive aggression. When individuals perceive threats, whether external or within their social group, the instinctive response may be to protect their interests by undermining others. This defensive aggression can manifest in various ways, from hostile behaviors in personal relationships to larger political or economic confrontations. Fear-driven actions are not limited to physical violence but can also involve psychological manipulation, eroding trust, and creating an atmosphere of suspicion. In competitive environments, individuals may resort to tactics like intimidation or disinformation to gain an advantage, further eroding cooperation within the group.

However, Halevy (2023) refrains from being all doom and gloom, suggesting that

fear-driven competition is not always maladaptive, but rather an evolutionary adaptation that may have helped protect resources and ensure individual survival in environments where competition for limited resources was common. Fear-induced competition can thus be an essential part of social dynamics, particularly within social hierarchies and power structures. When individuals or groups fear losing status, wealth, or influence, they may engage in competitive behaviors to maintain their position. This competition is observable in both human and non-human animal societies, where fear of social exclusion or domination fosters power-seeking behaviors aimed at securing and maintaining dominance.

The fear of powerlessness and subjugation plays a significant role in shaping group dynamics. This fear of being dominated or oppressed can drive individuals to assert their dominance or align with more powerful groups to avoid marginalization. For instance, in political and economic contexts, fear of disenfranchisement often leads individuals to form alliances with powerful factions, reinforcing existing power imbalances. In these cases, fear can fuel social inequality, with certain groups using the fear of others to maintain control over resources, status, and decision-making processes. This creates a cycle where fear and power are linked, with individuals using fear to protect and perpetuate their dominance.

However, fear-driven competition is not always marked by direct aggression. It can also appear as passive forms of social avoidance or exclusion. Individuals may fear being ostracized or rejected by the group and, as a result, engage in behaviors that distance themselves from others to avoid conflict. This type of social avoidance is common in tightly-knit communities or competitive work environments, where the fear of failing to meet expectations leads to withdrawal, reduced collaboration, and disengagement. The outcome is a fragmented social landscape, where cooperation is undermined not by open hostility but by the subtle, powerful

dynamics of fear-based exclusion.

The competitive and defensive dimensions of fear also affect collective decision-making. Research by Chang et al. (2022) shows that fear can cause individuals to prioritize self-interest over the collective good, especially in uncertain or resource-scarce situations. In these contexts, fear of being left behind or marginalized may lead individuals to act in ways that focus on their own survival rather than the success of the group. Fear-based individualism thus poses challenges for collective problem-solving and cooperation, particularly when the fear of losing status, resources, or power takes precedence over the benefits of collaboration.

Deriving Pleasure from Fear

Surely, for fear to have been and continue to be so prevalent in humanity's psyche, it must be pleasurable in some contexts. Who doesn't like taking a risk every now and then? Andersen et al.'s (2020) research answers that question. Their field study on recreational horror experiences revealed a complex relationship between fear and enjoyment, characterized by an inverted U-shaped curve of psychological response. This curve shows that mild to moderate fear can increase enjoyment, while excessive fear may overwhelm the psychological system, leading to discomfort or disengagement.

With balancing and controlling fear, humans have developed the capacity to modulate and reframe fear experiences, turning them from purely threatening stimuli into sources of psychological stimulation and pleasure. The ability to willingly seek out fear-inducing experiences, such as watching horror films, engaging in extreme sports, or breaking into an abandoned house, demonstrates the flexibility of human emotional processing. By interacting with fear in controlled environments, individuals can experience heightened arousal and reward,

which allows for emotional release and personal growth. This ability to manage fear in a safe context provides a sense of control and mastery, explaining the appeal of fear-based activities.

Additionally, recreational engagement with fear serves as a form of psychological exploration, giving individuals the chance to confront and process their anxieties in a low-risk setting. These experiences can also increase emotional resilience, helping individuals build tolerance to stress and fear over time. Only humans have the luxury to voluntarily interact with fear as for other animals fear is based in survival and therefore isn't something to fool around with. These luxuries are what set humans apart in degree rather than uniqueness, giving them greater "comfort" within Nature.

Existential and Philosophical Dimensions

The above sections have all failed to answer the first question: What is fear? We may never have a definite answer but for now philosophy may have a clue or two. Freud's (1930) conceptualization of fear offers a philosophical framework that goes beyond immediate psychological experiences. Fear, in his view, includes multiple dimensions: physiological decay, natural forces, intrapsychic conflicts, and existential anxiety. He suggests that fear is not just a biological reaction to external threats but an intrinsic part of the human condition, connected to our awareness of mortality, the fragility of existence, and our subconscious struggles with repressed desires.

This broad perspective drives home the point that fear is more than a survival mechanism, it is a core element of human psychological experience that shapes how we understand existence, mortality, and social relationships. Fear becomes a lens – a filter to our sensations and perceptions – through which humans assess their place in a complex and

unpredictable world. The awareness of life's impermanence, tied to our fears, drives many existential quests, such as the search for meaning, personal identity, and legacy. As humans face the inevitability of death, fear compels us to define the limits of our existence, often leading to philosophical and spiritual contemplation.

Fear's connection to intrapsychic conflict adds another layer to its philosophical importance. According to Freud, the tension between the conscious and unconscious mind is a constant source of anxiety, and this internal struggle can manifest as both fear of external events and fear of one's own internal desires or repressed thoughts. This conflict is central to many existential dilemmas, as it forces individuals to confront the darker aspects of their psyche, including fear of self-doubt, failure, and the unknowable nature of human consciousness.

In this way, fear is not only a response to external threats but a trigger for self-reflection and the formation of personal and collective meaning. It pushes humans to seek understanding of their place in the universe and their relationships with others. Through philosophy, religion, or art, humanity has tried to wrestle with the terror of the unknown and the certainty of death. Fear, in its various forms, shapes the existential dimensions of human life, acting as both a source of distress and a motivator for philosophical and spiritual exploration.

Conclusions

This review has examined the role of fear across human evolution, social dynamics, and individual psychology, both in humans and non-human animals. While often viewed negatively, fear is a powerful motivator that can drive cooperation, innovation, and social learning. By exploring fear through various disciplines, from evolutionary biology to cultural anthropology, we gain a clearer understanding of its role in shaping the human experience.

A persistent theme in the literature is the interaction between fear, cognition, and social behavior. Fear triggers physiological responses while also activating cognitive processes that influence how we perceive, interpret, and respond to the world. This interplay is evident in the development of social structures, cultural systems, and emotional responses. While fear is obviously present in non-human animals, it is primarily linked to survival rather than higher-level cognitive processes. However, just as fear arbitrates our reality one can assume it does the same for non-human animals. Differences in degree are not limited to the phenomena of fear, they exist on our entire concepts of reality. The debate of human uniqueness, therefore, is largely trite in practice. The differences in realities make it so we cannot accurately compare human traits with non-human traits. Non-human fear fits their reality just as human fear fits ours and attempting to debate or prove our uniqueness is destructive rather than productive to life as a whole.

A critical area for future research is the intersection of fear and emerging technologies. As we increasingly rely on technology for communication, information, and social interaction, unintended consequences are becoming more apparent. The constant stream of information, amplified by algorithms, fosters passive overconsumption and a lack of agency. This could create a feedback loop, where the offloading of cognitive functions to technology makes us more vulnerable to fear and manipulation. As a result, individuals may seek protection and cooperation within technological systems, further increasing reliance on them.

Understanding the relationship between fear, technology, and human cognition is vital for addressing the challenges ahead. I hope to continue exploring this in graduate school, developing strategies to mitigate its negative impacts while leveraging its potential for positive social change.

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